

Competitive Programming Notebook

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BIT MANIPULATION

Binary representation

```
1 string tobit(long long x) {
2     string a(64, '0');
3     for (int i = 63; i >= 0; --i) {
4         if (x & 1) a[i] = '1';
5         x >>= 1;
6     }
7     return a;
8 }
```

DATA STRUCTURES

Disjoint set union

```
1 const int N = 1e5;
2 struct DSU{
3     int num;
4     vector<int> parent;
5     vector<int> sizes;
6     DSU(){};
7     DSU(int n){
8         init(n);
9     }
10    void init(int n){
11        num = n;
12        parent.resize(n+1);
13        sizes.resize(n+1);
14        for(int i = 1; i <= n ; i++){
15            parent[i] = i;
16            sizes[i] = 1;
17        }
18    }
19    int find(int a){
20        if(a == parent[a]) return a;
21        return parent[a] = find(parent[a]);
22    }
23    void join(int a, int b){
24        int x = find(a);
25        int y = find(b);
26        if(x == y) return;
27        num--;
28        if(sizes[x] < sizes[y]){
29            parent[x] = y;
30            sizes[y] += sizes[x];
31        }
32        else{
33            parent[y] = x;
34            sizes[x] += sizes[y];
35        }
36    }
37 }
38 ;;
```

Segment tree

```
1 const int N = 1e5;
2 struct info{
3     int num;
4 };
5 info ST[4*N];
6 info ope(info u, info v){
7     return {u.num+v.num};
8 }
9 void build(int in, int L, int R){
10    if(L == R){
11        cin >> ST[in].num;
12    }
13    else{
14        int mid = (L+R)>>1;
15        build((2*in),L,mid);
16        build((2*in)+1,mid+1,R);
17        ST[in] = ope(ST[(2*in)], ST[(2*in)+1]);
18    }
19 }
20 void update(int in, int L, int R, int pos, int val){
```

```

21     if(L == R){
22         ST[in].num = val;
23     }
24     else{
25         int mid = (L+R)>>1;
26         if(pos <= mid){
27             update(2*in,L,mid,pos,val);
28         }
29         else{
30             update((2*in)+1,mid+1,R,pos,val);
31         }
32         ST[in] = ope(ST[2*in], ST[(2*in)+1]);
33     }
34 }
35 info get(int in, int L, int R, int l, int r){
36     if(r < L or R < l) return {0};
37     if(l <= L and R <= r) return ST[in];
38     int mid = (L+R)>>1;
39     info left = get(2*in,L,mid,l,r);
40     info right = get((2*in)+1,mid+1,R,l,r);
41     return ope(left,right);
42 }
43 // v = {1 2 3 4 5}
44 // get(1, 0, n-1, 1, 3) = 2 + 3 + 4 = 9

```

Sparse table

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int v[N]; //values
4  int st[N][LOG]; //Sparse Table
5  int lg[N]; // log values
6  int n;
7  void build(){
8      for(int i = 0; i < n; i++) st[i][0] = v[i];
9
10     for(int i = 1; i < LOG; i++){
11         for(int j = 0; j < n - (1<<i) + 1; j++){
12             st[j][i] = min(st[j][i-1],st[j+(1<<(i-1))][i-1]);
13         }
14     }
15     lg[1] = 0;
16     for(int i = 2; i <= n;i++) lg[i] = lg[i/2] + 1;
17 }
18 int query(int L, int R){
19     int k = lg[R-L+1];
20     return min(st[L][k],st[R-(1<<k)+1][k]);
21 }
22 // v = [5, 2, 4, 7, 1, 3]
23 // query(1, 4) [2, 4, 7, 1]

```

FLOW

Dinic

```

1  struct Dinic
2  {
3      const int INF = 1e9;
4
5      struct flowEdge
6      {
7          int to, rev, f, cap;
8      };
9
10     vector<vector<flowEdge>> G;
11     vector<int> work, lvl, Q;
12
13     int S, T, nodes;
14
15     Dinic(){}
16     Dinic(int n, int s, int t){
17         S = s;
18         T = t;
19         nodes = n;
20         G.clear();
21         G.resize(n);
22         Q.resize(n);
23     }
24     void addEdge(int st, int en, int cap){
25         flowEdge A = {en, (int)G[en].size(), 0, cap};

```

```

26     flowEdge B = {st, (int)G[st].size(), 0, 0};
27     G[st].pb(A);
28     G[en].pb(B);
29 }
30 int dfs(int v, int f){
31     if(v == T or f == 0) return f;
32     for(int &i = work[v]; i < G[v].size(); i++){
33         flowEdge &e = G[v][i];
34         int u = e.to;
35         if(e.cap <= e.f or lvl[u] != lvl[v] + 1) continue;
36         int df = dfs(u, min(f, e.cap - e.f));
37         if(df){
38             e.f += df;
39             G[u][e.rev].f -= df;
40             return df;
41         }
42     }
43     return 0;
44 }
45 bool bfs(){
46     int qt = 0;
47     Q[qt++] = S;
48     lvl.assign(nodes, -1);
49     lvl[S] = 0;
50     for(int i = 0; i < qt; i++){
51         int u = Q[i];
52         for(flowEdge &e : G[u]){
53             int v = e.to;
54             if(e.cap <= e.f or lvl[v] != -1) continue;
55             lvl[v] = lvl[u] + 1;
56             Q[qt++] = v;
57         }
58     }
59     return lvl[T] != -1;
60 }
61 int maxFlow(){
62     int flow = 0;
63     while(bfs()){
64         work.assign(nodes, 0);
65         while(true){
66             int df = dfs(S, INF);
67             if(df == 0) break;
68             flow += df;
69         }
70     }
71     return flow;
72 }
73
74 };

```

GEOMETRY

Convex hull

```

1 // Devuelve sentido horario
2 vector<point> hull(vector<point> p)
3 {
4     int n = p.size();
5     vector<point> h;
6     sort(p.begin(), p.end());
7     for(int i = 0; i < n; i++){
8         while(h.size() >= 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
9             h.pop_back();
10        }
11        h.push_back(p[i]);
12    }
13    h.pop_back();
14    int k = h.size();
15    for(int i = n-1; i >= 0; i--){
16        {
17            while(h.size() >= k + 2 and p[i].left(h[h.size() - 2], h.back()) < 0){
18                h.pop_back();
19            }
20            h.pb(p[i]);
21        }
22        h.pop_back();
23    }
24 }

```

Point

```

1 struct point
2 {
3     int x, y;
4     point() {}
5     point(int x, int y): x(x), y(y) {}
6     point operator -(point p) {return point(x - p.x, y - p.y);}
7     point operator +(point p) {return point(x + p.x, y + p.y);}
8     int sq() const {return x * x + y * y;}
9     double abs() const {return sqrt(sq());}
10    int operator ^(point p) {return x * p.y - y * p.x;}
11    int operator *(point p) {return x * p.x + y * p.y;}
12    point operator *(int a) {return point(x * a, y * a);}
13    bool operator <(const point& p) const {return x == p.x ? y < p.y : x < p.x;}
14    int left(point a, point b) {
15        return ((b - a) ^ (*this - a));
16    }
17 };
18 ostream& operator<<(ostream& os, const point& p) {
19     return os << "(" << p.x << "," << p.y << ")\n";
20 }
21
22
23 void polarSort(vector<point>& v) {
24     sort(v.begin(), v.end(), [] (point a, point b) {
25         const point origin{0, 0};
26         bool ba = a < origin, bb = b < origin;
27         if (ba != bb) { return ba < bb; }
28         return (a^b) > 0;
29     });
30 }

```

GRAPH ALGORITHMS

Bfs

```

1 const int N = 1e5;
2
3 vector<int> G[N];
4 int vis[N];
5 void bfs(int source){
6     queue<int> q;
7     q.push(source);
8     vis[source] = 0;
9     while(!q.empty()){
10        int u = q.front();
11        q.pop();
12        for(int v : G[u]){
13            if(vis[v] == -1){
14                vis[v] = vis[u] + 1;
15                q.push(v);
16            }
17        }
18    }
19 }

```

Dfs

```

1 const int N = 1e5;
2
3 vector<int> G[N];
4 int vis[N];
5
6 void dfs(int u){
7     vis[u] = 1;
8     for(int v : G[u]){
9         if(vis[v] == -1){
10            dfs(v);
11        }
12    }
13 }

```

Dijkstra

```

1 const int N = 1e5;
2
3 int vis[N];

```

```

4 vector<pair<int,int>> G[N];
5
6 void dijkstra(int x){
7     priority_queue<pair<int,int>, vector<pair<int,int>>, greater<pair<int,int>>> pq;
8     pq.push({0,x});
9     vis[x] = 0;
10    while(!pq.empty()){
11        auto y = pq.top();
12        pq.pop();
13        int u = y.second;
14        int cost = y.first;
15        for(auto v : G[u]){
16            int cur = cost + v.second;
17            if(cur < vis[v.first]){
18                vis[v.first] = cur;
19                pq.push({cur,v.first});
20            }
21        }
22    }
23 }

```

Kruskal mst

```

1 const int N = 1e5;
2 struct DSU{
3     int num;
4     vector<int> parent;
5     vector<int> sizes;
6     DSU(){};
7     DSU(int n){
8         init(n);
9     }
10    void init(int n){
11        num = n;
12        parent.resize(n+1);
13        sizes.resize(n+1);
14        for(int i = 1; i <= n ; i++){
15            parent[i] = i;
16            sizes[i] = 1;
17        }
18    }
19    int find(int a){
20        if(a == parent[a]) return a;
21        return parent[a] = find(parent[a]);
22    }
23    void join(int a, int b){
24        int x = find(a);
25        int y = find(b);
26        if(x == y) return;
27        num--;
28        if(sizes[x] < sizes[y]){
29            parent[x] = y;
30            sizes[y] += sizes[x];
31        }
32        else{
33            parent[y] = x;
34            sizes[x] += sizes[y];
35        }
36    }
37 };
38 int n;
39 vector<pair<int,pair<int,int>>> List;
40 int Kruskal_MST(){
41     sort(List.begin(),List.end());
42     DSU dsu(n);
43     int mst = 0, t = 1;
44     for(int i = 0; i < List.size(); i++){
45         int w = List[i].first; //weight
46         int a = List[i].second.first; //at
47         int b = List[i].second.second; //to
48         if(dsu.find(a) != dsu.find(b)){
49             dsu.join(a,b);
50             mst += w;
51             t++;
52         }
53         if(t == n) break;
54     }
55     return mst;
56 }

```

Topological sort kahn algorithm


```

1  const int N = 1e5;
2
3  int indegree[N];
4  vector<int> G[N];
5  int n;
6  vector<int> Kahn(){
7      queue<int> q;
8      vector<int> ans;
9      for(int i = 0; i < n; i++){
10         if(indegree[i] == 0) q.push(i);
11     }
12     while (!q.empty()){
13         int v = q.front();
14         q.pop();
15         ans.push_back(v);
16         for(int u : G[v]){
17             indegree[u]--;
18             if(indegree[u] == 0){
19                 q.push(u);
20             }
21         }
22     }
23     //if(ans.size() != n) grafo no DAG
24     return ans;
25 }

```

MATH

Binary exponentiation

```

1  long long pote(long long a, long long b, long long mod){
2      long long ans = 1ll;
3      while(b){
4          if(b&1) ans = (ans * a)%mod;
5          a = (a * a)%mod;
6          b>>=1;
7      }
8      return ans;
9  }
10 // a^-1 = a^mod-2 = pote(a, mod-2);

```

Modular inverse

```

1  const long long N = 1e6 + 10;
2  const int MOD = 1e9+7;
3  long long fact[N];
4  long long invfact[N];
5  void init(){
6      fact[0] = 1;
7      for(long long i = 1; i < N; i++){
8          fact[i] = fact[i-1] * i % MOD;
9      }
10     invfact[N-1] = pote(fact[N-1], MOD-2);
11
12     for(long long i = N-2; i >= 0; i--){
13         invfact[i] = invfact[i+1] * (i+1) % MOD;
14     }
15
16     long long nCk(long long n, long long k){
17         if(k < 0 || k > n) return 0;
18         return fact[n] * invfact[k] % MOD * invfact[n-k] % MOD;
19     }

```

Pollard rho rabin miller

```

1  map<long long, long long> fact;
2
3  long long mcd(long long a, long long b) {
4      a = abs(a);
5      b = abs(b);
6      while (a != 0) {
7          long long r = b % a;
8          b = a;
9          a = r;
10     }
11     return b;

```

```

12 }
13 //Rabin Miller
14 bool primo(long long n) {
15     if (n < 2) return false;
16     if (n == 2) return true;
17     long long D = n - 1, S = 0;
18     while (D % 2 == 0) {
19         D /= 2;
20         S++;
21     }
22     static const long long STEPS = 1611;
23     for(long long pasos = 0; pasos < STEPS; pasos++){
24         const long long A = 1 + rand() % (n - 1);
25         long long M = pote(A, D, n);
26         if (M == 1 || M == (n - 1)) goto next;
27         for(long long k = 0; k < S-1; k++){
28             M = ((M) * M) % n;
29             if (M == (n - 1)) goto next;
30         }
31         return false;
32     next:;
33     }
34     return true;
35 }
36
37 //Rho de pollard
38 long long factor(long long n) {
39     static long long A, B;
40     A = 1 + rand() % (n - 1);
41     B = 1 + rand() % (n - 1);
42     #define fun(x) (((x) * (x + B)) % n + A)
43
44     long long x = 2, y = 2, d = 1;
45     while (d == 1 || d == -1) {
46         x = fun(x);
47         y = fun(fun(y));
48         d = mcd(x - y, n);
49     }
50     return abs(d);
51 }
52
53 void factorize(long long n){
54     assert(n > 0);
55     while (n > 1 && !primo(n)) {
56         long long fx;
57         do {
58             fx = factor(n);
59         } while (fx == n);
60
61         n /= fx;
62         factorize(fx);
63
64         for (auto &it : fact) {
65             while (n % it.first == 0) {
66                 n /= it.first;
67                 it.second++;
68             }
69         }
70     }
71     if (n > 1) fact[n]++;
72 }

```

Sieve of eratosthenes

```

1  const int N = 1e7;
2  int sieve[N];
3  void make(){
4      for(int i = 0; i < N; i++) sieve[i] = 1;
5      sieve[0] = sieve[1] = 0;
6      for (int i = 2; i < N; i++)
7      {
8          if(sieve[i]){
9              for (int j = i*i; j < N; j+=i)
10                 {
11                     sieve[j] = 0;
12                 }
13             }
14     }
15 }

```

STRINGS

Hashing

```

1 struct Hash {
2     long long P=1777771,MOD[2],PI[2];
3     vector<long long> h[2],pi[2];
4     Hash(string& s){
5         MOD[0]=999727999;MOD[1]=107077777;
6         PI[0]=325255434;PI[1]=10018302;
7         for(int k = 0; k < 2; k++) h[k].resize(s.size()+1),pi[k].resize(s.size()+1);
8         for(int k = 0; k < 2; k++){
9             h[k][0]=0;pi[k][0]=1;
10            long long p=1;
11            for(int i = 1; i < s.size()+1; i++){
12                h[k][i]=(h[k][i-1]+p*s[i-1])%MOD[k];
13                pi[k][i]=(1LL*pi[k][i-1]*PI[k])%MOD[k];
14                p=(p*P)%MOD[k];
15            }
16        }
17    }
18    long long get(int s, int e){
19        long long h0=(h[0][e]-h[0][s]+MOD[0])%MOD[0];
20        h0=(1LL*h0*pi[0][s])%MOD[0];
21        long long h1=(h[1][e]-h[1][s]+MOD[1])%MOD[1];
22        h1=(1LL*h1*pi[1][s])%MOD[1];
23        return (h0<<32)|h1;
24    }
25 };

```

Kmp

```

1 vector<int> kmp(string s){
2     int n = s.size();
3     vector<int> pi(n);
4     for(int i = 1; i < n; i++){
5         int j = pi[i-1];
6         while(j > 0 and s[i] != s[j]){
7             j = pi[j-1];
8         }
9         if(s[i] == s[j]){
10            j++;
11        }
12        pi[i] = j;
13    }
14    return pi;
15 }
16 //Pattern Matching
17 vector<int> ocurrencia(string texto, string patron) {
18     vector<int> P = kmp(patron);
19     int k = 0;
20     vector<int> M;
21     repl(i,0,texto.size()) {
22         while (k > 0 and patron[k] != texto[i]) k = P[k - 1];
23         if (patron[k] == texto[i]) k++;
24         if (k == patron.size()) {
25             M.pb(i - k + 1);
26             k = P[k - 1];
27         }
28     }
29     return M;
30 }
31
32 //Automata
33 void genAut(string &s) {
34     int n = s.size();
35
36     vector<vector<int>> aut(n+1, vector<int>(26));
37     vector<int> pi = kmp(s);
38
39     for(int i = 0 ; i < n; i++){
40         aut[i][s[i]-'a'] = i+1;
41     }
42
43     for(int i = 0; i < n+1; i++){
44         for(int c = 0; c < 26; c++){
45             if (aut[i][c]) continue;
46             if (i < n and s[i] == 'a' + c) aut[i][c] = i+1;
47             else if (i > 0) aut[i][c] = aut[pi[i-1]][c];
48             else aut[i][c] = 0;
49         }
50     }
51 }

```

TREE ALGORITHMS

Binary lifting

```

1  const int N = 1e5;
2  const int LOG = 20;
3  int up[N][LOG];
4  vector<int> G[N];
5  //Find parents
6  void dfs(int u){
7      for(int v : G[u]){
8          up[v][0] = u;
9          for(int j = 1; j < LOG; j++){
10             up[v][j] = up[up[v][j-1]][j-1];
11         }
12         dfs(v);
13     }
14 }
15 int get_K_parent(int a, int k){
16     for(int j = LOG-1; j >= 0; j++){
17         if(k & (1<<j)){
18             a = up[a][j];
19         }
20     }
21     return a;
22 }

```

Euler tour

```

1  const int N = 1e5 + 10;
2  vector<int> path;
3  vector<int> G[N];
4  void EulerTour(int node, int parent){
5      path.pb(node);
6      for(int u : G[node]){
7          if(u != parent){
8              EulerTour(u,node);
9          }
10     }
11     path.pb(node);
12 }
13
14 const int N = 1e5 + 10;
15 int tin[N];
16 int tout[N];
17 int flat[N];
18 int timer = 1;
19 void EulerTour(int node, int parent){
20     tin[node] = timer;
21     flat[timer] = node;
22     timer++;
23     for(int u : G[node]){
24         if(u != parent){
25             EulerTour(u, node);
26         }
27     }
28     tout[node] = timer;
29 }

```

Lowest common ancestor

```

1  const int N = 1e5;
2  const int LOG = 20;
3  vector<int> G[N];
4  int depth[N];
5  int up[N][LOG];
6  // Assign levels and get parents
7  void dfs(int u){
8      for(int v : G[u]){
9          depth[v] = depth[u] + 1;
10         up[v][0] = u;
11         for(int j = 1; j < LOG; j++){
12             up[v][j] = up[up[v][j-1]][j-1];
13         }
14         dfs(v);
15     }
16 }
17 int get_lca(int a, int b){
18     if(depth[a] < depth[b]) swap(a,b);
19     int k = depth[a] - depth[b];

```

```
20     for(int j = LOG-1; j >= 0; j--){
21         if(k & (1<<j)){
22             a = up[a][j];
23         }
24     }
25     if(a == b) return a;
26     for(int j = LOG-1; j >= 0; j--){
27         if(up[a][j] != up[b][j]){
28             a = up[a][j];
29             b = up[b][j];
30         }
31     }
32     return up[a][0];
33 }
```